

Notes: Juhász, Squicciarini, and Voigtländer (JPE, 2024)

Technology Adoption and Productivity Growth:

Evidence from Industrialization in France

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1 Big picture

The paper studies *why measured productivity often rises slowly after a major technology arrives*. The central hypothesis is that adopting a new *technology* is not enough: firms must also learn *how to reorganize production* to use the technology effectively. Because such reorganization is complex, early adopters experiment and make mistakes, generating large initial dispersion in productivity. Over time, firms learn about “best practice” and diffuse this knowledge across firms and space, producing a distinctive (and testable) pattern of productivity growth.

Core empirical setting: plant-level labor productivity (output per worker) in early industrial France, comparing a major “new-technology + reorganization” sector (cotton spinning) to two more mature sectors (metallurgy and paper milling), around 1800 vs. 1840.

2 Historical background and why cotton spinning is special

2.1 Industrial Revolution in Britain: mechanization plus coordination

Mechanized cotton spinning (spinning jenny, water frame/throstle, mule) was one of the hallmark technologies of the Industrial Revolution. A key feature is that mechanization *changed the optimal organization of production*: the technology was most effective in factories where tasks were synchronized, workers were disciplined, and machines were maintained and powered reliably.

2.2 Cotton spinning in France: a “reorganization” problem

In France, the technology arrived later, and early adopters faced a hard organizational problem: they had to learn how to redesign workflow, factory layout, and supervision to operate a mechanized plant efficiently. This creates scope for:

- **high initial productivity dispersion** (different organizational blueprints),
- **high exit** (failed experiments),
- **spillovers** (imitation / learning from successful plants).

2.3 Comparison sectors: metallurgy and paper milling

Metallurgy and paper milling provide benchmarks where factory organization and production methods are closer to “best practice” already by 1800. If the key friction is *learning how to reorganize around a new technology*, productivity dynamics in cotton spinning should look different.

3 Data and measurement

3.1 Plant-level surveys and repeated cross-sections

The paper uses historical industrial surveys with plant-level information on: location, employment, output value, and (for cotton spinning) machinery and plant characteristics. The data provide repeated cross-sections around 1800 and again around 1840.

3.2 Productivity measure

For plant i in sector s and period t , define labor productivity as

$$\text{Prod}_{ist} \equiv \frac{Y_{ist}}{L_{ist}}, \quad \ln \text{Prod}_{ist} = \ln Y_{ist} - \ln L_{ist}, \quad (1)$$

where Y_{ist} is real output (after deflation and quality adjustments) and L_{ist} is employment. The paper stresses that careful output measurement matters in historical data, and reports robustness checks.

4 Empirical fact: productivity growth is “lower-tail driven” in cotton spinning

4.1 Average and quantile productivity growth: basic model

A reduced-form way to summarize productivity change is to pool both periods and estimate:

$$\ln \text{Prod}_{ist} = \alpha_s + \delta_s \cdot \mathbf{1}\{t = 1840\} + u_{ist}. \quad (2)$$

Here δ_s is total log growth from the early period to 1840 (annualized by dividing by the number of years). To study distributional change, estimate the same specification by *quantile regression* at quantile q :

$$Q_q(\ln \text{Prod}_{ist} \mid \mathbf{1}\{t = 1840\}) = \alpha_s(q) + \delta_s(q) \cdot \mathbf{1}\{t = 1840\}. \quad (3)$$

4.2 Table 1: how to read it and what it says

Table 1 reports annualized growth rates (in percent) at different quantiles.

- **Cotton spinning:** strong catch-up from below. The 10th percentile grows at about 3.92% per year, while the 90th percentile grows at about 1.01% per year (monotone decline across quantiles).
- **Metallurgy:** growth is not lower-tail dominated (upper quantiles grow relatively fast; e.g. $\approx 3.00\%$ at the 90th percentile).
- **Paper milling:** modest, fairly flat growth (below 1% per year across quantiles).

Interpretation: in cotton spinning, dispersion shrinks mainly because the bottom improves (or disappears), not because the frontier moves a lot.

4.3 Figures 1–2: distributional signature of reorganization learning

Figure 1 shows the productivity distributions at the beginning vs. end of the sample period. For cotton spinning, the initial distribution is very dispersed and the *lower tail largely disappears* by 1840. Figure 2 visualizes the quantile-growth profile: cotton spinning displays a steep downward slope across quantiles, consistent with lower-tail catch-up.

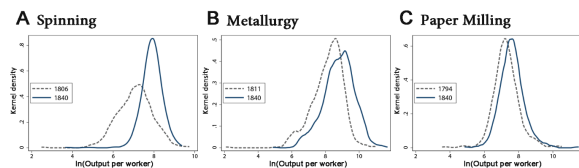


FIG. 1.—Changes in the productivity distributions in the three sectors. The figure shows the distribution of log(output per worker) for the three sectors at the beginning of our sample period (around 1800) and in 1840. Productivity growth in spinning was mainly due to a disappearing lower tail. In contrast, in metallurgy and paper milling, the whole distribution shifted to the right.

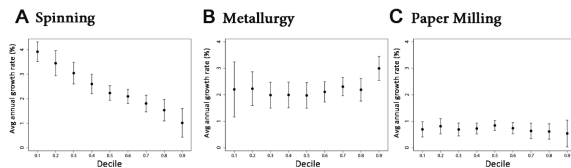


FIG. 2.—Productivity growth at different quantiles of the distribution. The figure visualizes the results of quantile regressions for growth in log(output per worker) for the three sectors, estimated at each decile. Productivity growth in spinning was concentrated in the lower tail of plant productivity. In contrast, in metallurgy and paper milling, productivity growth occurred relatively evenly across the distribution.

4.4 Why lower-tail growth points to “best-practice diffusion”

A pure “frontier technology improvement” story would often shift the whole distribution (or expand the upper tail). The cotton-spinning pattern is closer to:

- best practice exists early (some plants are already highly productive),
- many plants initially operate far below best practice (organizational mistakes),
- over time, these plants either exit or learn/improve, compressing dispersion from below.

5 Mechanism framework: learning about best practice (stylized model)

5.1 Complementarities across tasks

The paper highlights that mechanized production involves multiple complementary tasks (machine operation, maintenance, power management, workflow coordination, supervision). With strong complementarities, a “bad” organizational choice in one task can depress the whole plant’s productivity.

5.2 A minimal dynamic “search and imitation” model

Plants differ in an organizational efficiency parameter a (“how well the factory is organized”). Each period, a plant chooses between:

- **Produce:** operate with current efficiency a ,
- **Search/copy:** temporarily forgo production to learn a better method from another producer,
- **Exit:** shut down.

A compact representation is a Bellman equation:

$$V(a) = \max \left\{ \underbrace{a + \beta V(a)}_{\text{produce}}, \underbrace{-c + \beta \mathbb{E}[V(\max\{a, a'\})]}_{\text{search and imitate}}, \underbrace{0}_{\text{exit}} \right\}, \quad (4)$$

where $c > 0$ is a search cost and a' is the efficiency of a randomly matched producer. In equilibrium, there are cutoffs:

$$a < \underline{a} \Rightarrow \text{exit}, \quad \underline{a} \leq a < \bar{a} \Rightarrow \text{search}, \quad a \geq \bar{a} \Rightarrow \text{produce}.$$

This generates a *truncated/compressed* distribution over time: inefficient plants exit or search and adopt better practice, shrinking the lower tail without requiring large frontier improvements.

5.3 Figure 3: visual summary

Figure 3 illustrates this mechanism: the initial distribution is wide; low-productivity mass disappears; the upper tail changes little.

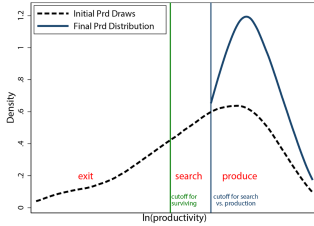


FIG. 3.—Productivity (Prd) dynamics in our stylized model. In the initial period, the least productive plants exit. Subsequently, in the innovation period, surviving plants decide whether to search for better organizational knowledge, forgoing production. The searching plants are randomly matched to continuing producers as in Perla and Tonetti (2014). In the final period, searching plants adopt the improved organizational knowledge from more productive producing plants. Thus, the lower tail of the productivity distribution disappears, and the mass shifts toward higher-productivity draws. At the same time, the productivity frontier remains unchanged.

TABLE 1
ANNUAL PRODUCTIVITY GROWTH (%) AT DIFFERENT QUANTILES OF THE DISTRIBUTION

	AT THE QUANTILES					
	AVERAGE	.1	.25	.5	.75	.9
	(1)	(2)	(3)	(4)	(5)	(6)
Spinning (1806–40)	2.420***	3.917***	3.295***	2.254***	1.651***	1.014***
	(.154)	(.204)	(.229)	(.151)	(.167)	(.257)
Metallurgy (1811–40)	2.328***	2.205***	2.068***	1.979***	2.285***	2.998***
	(.183)	(.530)	(.317)	(.247)	(.193)	(.232)
Paper milling (1794–1840)	.719***	.697***	.717***	.846***	.691***	.542***
	(.111)	(.145)	(.139)	(.092)	(.130)	(.258)
						867

NOTE.—The table reports the average annual productivity growth between the initial sample period (around 1800) and 1840 for the three sectors (col. 1), and annual productivity growth estimated at different quantiles (cols. 2–6). Column 7 reports the number of observations. Robust standard errors are in parentheses.

*** $p < .01$.

TABLE 2
SURVIVAL RATES ACROSS SECTORS

Period	Spinning 1806–40	Metallurgy 1811–40	Paper Milling 1794–1840
Baseline sample:			
Survival rate (%)	5.0	37.7	10.8
Number of plants	340	470	520
Restricted sample:			
Survival rate (%)	12.6	52.5	24.6
Number of plants	87	324	211

NOTE.—“Survival rate” is defined as the percentage of plants from the initial period that survived to the later period, based on matching by either name or location (see sec. III.B). The restricted-sample survival rate uses the subset of plants located in communes that had only one plant in the initial period.

6 Evidence consistent with “learning about best practice”

6.1 Evidence 1: high exit in cotton spinning (Table 2)

A strong prediction of experimentation is a shakeout. Table 2 reports survival rates from the early survey to 1840.

- **Cotton spinning:** about 5% survival in the baseline sample (and 12.6% in a conservative restricted sample).
- **Metallurgy:** about 37.7% survival (baseline).

- **Paper milling:** about 10.8% survival (baseline).

Interpretation: cotton spinning looks like an early-stage industry with many failed organizational experiments.

6.2 Evidence 2: younger plants are *more* productive (Table 3)

A pure within-plant learning-by-doing story often predicts older plants are more productive. Instead, in cotton spinning (1806), young plants outperform older plants.

6.2.1 Regression model

For cotton spinning plants in 1806:

$$\ln \text{Prod}_i = \beta_0 + \beta_1 \mathbf{1}\{\text{Young}_i\} + X_i' \gamma + \varepsilon_i, \quad (5)$$

where $\mathbf{1}\{\text{Young}_i\}$ indicates plants younger than the sample median age, and X_i includes controls such as spindles per worker, plant size, and machine-type (vintage) indicators.

TABLE 3
COTTON SPINNING IN 1806: PRODUCTIVITY AND PLANTS' AGE PROFILE ($N = 340$)

	Dependent Variable: $\log(\text{Output per Worker})$					
	(1)	(2)	(3)	(4)	(5)	(6)
Young plant	.575*** (.088)	.534*** (.085)	.608*** (.086)	.543*** (.083)	.575*** (.089)	.493*** (.085)
$\log(\text{Spindles/worker})$		.336*** (.070)				
$\log(\text{Workers})$			.107*** (.025)			
Spinning jenny				-.626*** (.087)		
Throstle					-.003 (.092)	
Mule jenny						.481*** (.086)
R^2	.11	.17	.14	.20	.11	.18

NOTE.—The table shows that mechanized cotton-spinning plants that had just entered the market by 1806 had significantly higher productivity. “Young plant” is a dummy variable equal to 1 for cotton-spinning plants with below-median age (with the median age in 1806 being 3 years). The number of spindles is a standard measure of a spinning machine’s production capacity, irrespective of vintage. “Spinning jenny,” “throstle,” and “mule jenny” are binary indicators equal to 1 for plants using the earliest (spinning jenny), intermediate (throstle, or water frame), and latest (mule jenny) vintage of spinning machinery, respectively. Robust standard errors are in parentheses.

*** $p < .01$.

6.2.2 Interpretation

Across specifications, $\beta_1 \approx 0.5$ – 0.6 . This implies young plants have on the order of 50–80% higher output per worker (depending on whether one uses the log approximation or exponentiation). Importantly, controlling for machine type does not eliminate the young-plant advantage: new entrants appear to benefit from accumulated *organizational knowledge* rather than just newer machines.

6.3 Evidence 3: convergence in factory design (Figure 4)

The paper documents convergence in physical layout (a proxy for organizational best practice). Two dimensions are:

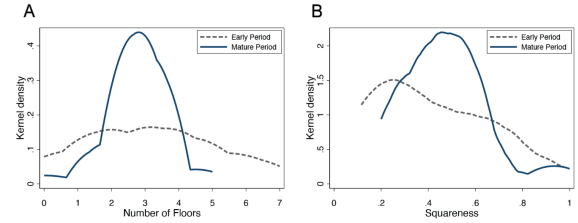


FIG. 4.—Experimentation and convergence in mill design for cotton-spinning plants. The figure shows the distributions of two important features of cotton mill design, in an “early” experimental period (before 1820) and a later “mature” period (after 1820): the number of floors in cotton mills (A) and “squareness,” as defined in section V.B (B). Data on the length, width, and number of floors of 59 cotton plants are from Chassagne (1991).

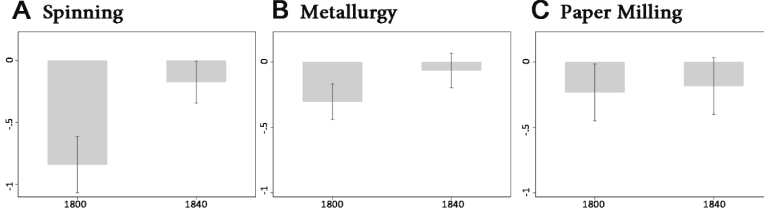


FIG. 5.—Proximity to high-productivity plants. The figure shows that proximity to high-productivity plants mattered the most in mechanized cotton spinning at the beginning of our sample period (around 1800), when the technology had just been introduced in France. The panels plot the standardized beta coefficients of $\ln \text{Dist}^{p90}$, which measures the log distance to the closest plant with productivity in the 90th percentile (in the same sector and in the same period—1800 and 1840, respectively). The dependent variable is $\log(\text{output per worker})$. All regressions include *département* fixed effects (see table A.10). Whiskers indicate 90% confidence intervals.

- **Number of floors:** early mills vary widely; later mills concentrate around a narrower mode.
- **Squareness:** define

$$S \equiv \frac{\text{length} \times \text{width}}{\max\{\text{length}, \text{width}\}^2} = \frac{\min\{\text{length}, \text{width}\}}{\max\{\text{length}, \text{width}\}} \in (0, 1], \quad (6)$$

where $S = 1$ is square and smaller S is more elongated.

Convergence supports the idea that firms learn a “best-practice blueprint” for organizing mechanized production.

6.4 Evidence 4: spatial diffusion of knowledge (Figure 5)

If best practices diffuse locally, plants closer to highly productive incumbents should be more productive.

6.4.1 Empirical specification

For plant i in geography unit j (e.g., a French *département*), define:

$\text{Dist}_{ij}^{p90} \equiv$ distance to the nearest same-sector plant with productivity in the 90th percentile.

Estimate:

$$\ln \text{Prod}_{ij} = \beta_0 + \beta_1 \ln \text{Dist}_{ij}^{p90} + \text{FE}_j + \varepsilon_{ij}. \quad (7)$$

6.4.2 Interpretation

A negative β_1 means that being farther from top performers is associated with lower productivity. Figure 5 reports standardized coefficients: in cotton spinning in the early period, the standardized relationship is large (around -0.84), and it weakens later and in the comparison sectors. This is consistent with geographically localized diffusion of best practice when the technology is new.

7 Short summary

- **Bottom line:** in the early phase of mechanized cotton spinning, productivity growth mainly reflects the disappearance of low-productivity plants and the diffusion of organizational best practices, rather than large improvements at the technological frontier.
- **Intuition:** a major technology can be easy to copy mechanically (buy the machine) but hard to use well (redesign factory organization). The latter requires experimentation, selection, imitation, and time.
- **Broader implication:** “productivity paradoxes” after general-purpose technologies are consistent with slow organizational learning.

8 Conclusion

8.1 Summary

The paper concludes that the productivity gains from a major new technology can take decades to materialize because adoption is not a single act (buying a machine) but a **multi-dimensional reorganization problem**. In early industrial France, mechanized cotton spinning exhibits rapid productivity growth that is **driven primarily by the disappearance of the lower tail** of the productivity distribution rather than by large movements in the frontier. This pattern is consistent with a process in which “best practice” exists early (some plants are already highly productive), but many adopters initially operate far below best practice due to organizational mistakes; over time, poorly performing plants either **exit** or **learn and converge** toward superior organizational blueprints.

In contrast, more mature sectors (metallurgy and paper milling) do not show the same lower-tail-driven convergence, supporting the interpretation that cotton spinning was a setting where the key friction was **learning how to organize production around a new technology**.

8.2 Core empirical takeaways

1. **Lower-tail catch-up dominates:** in cotton spinning, productivity growth is steeply decreasing across quantiles, implying that low-productivity plants improve (or vanish) much more than high-productivity plants.
2. **High experimentation and selection:** cotton spinning shows extremely low survival rates from the early period to 1840, consistent with widespread experimentation and subsequent shakeout.
3. **Young plants outperform old plants:** within early cotton spinning, younger plants are more productive even after controlling for relevant observables. This pattern is difficult to rec-

oncile with pure within-plant learning-by-doing and instead suggests **cohort learning**: later entrants benefit from accumulated knowledge about how to organize mechanized production.

4. **Convergence in “factory blueprint”**: plant design characteristics converge over time (layout/floors/squareness), consistent with the diffusion of a best-practice organizational template.
5. **Spatial diffusion of knowledge**: proximity to high-productivity producers predicts higher productivity early on, and this relationship weakens later as knowledge diffuses more broadly.

8.3 Economic intuition (why the results make sense)

The central intuition is that mechanized production features **strong complementarities** across tasks: machine operation, maintenance, power management, workflow coordination, supervision, and worker discipline. When complementarities are strong, a single weak link (a poor organizational choice) can sharply reduce overall productivity. Hence early adopters of a new technology face a difficult search problem over organizational methods, producing:

- **large initial dispersion** (many organizational experiments),
- **high exit** (failed experiments are selected out),
- **imitation/diffusion** (plants learn from successful blueprints, especially locally),
- **distributional convergence from below** (the worst plants improve or disappear).

Thus, measured “technology-driven” productivity growth is often the outcome of **organizational learning and selection**, not only frontier innovation.

8.4 Broader implication

A key conceptual lesson is that productivity growth after new technologies depends on the accumulation and diffusion of **intangible organizational capital**. Therefore, policies or shocks that accelerate adoption may have limited short-run productivity effects if they do not also facilitate (i) organizational experimentation, (ii) managerial know-how diffusion, and (iii) the reallocation/exit process that removes persistently inefficient producers.

8.5 Takeaway

In early industrial France, mechanized cotton spinning shows that productivity growth after a major technology arrives is mainly a process of organizational learning and selection that eliminates the lower tail of productivity, rather than a rapid shift of the technological frontier.